Cloud Infrastructures

08 - The Google Cloud Platform

Juan Ángel Lorenzo del Castillo

CY Cergy Paris Université

ING3 - Ingénierie du Cloud Computing

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juan-angel.lorenzo-del-castillo@cyu.fr

CY TECH

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The Google Cloud Platform

- The Google Cloud Platform (GCP) was originally built to power, ingest, manage and serve all data from Google's own applications (Gmail, Google Maps, Chrome, YouTube, Android, Play, Drive).
- Numerous big data and Machine Learning (ML) products are built on top of this infrastructure.
- GCP provides lots of compute power to run big data jobs.
- Google trains its production ML models on its network of data centres and then deploys smaller trained versions of these models to the hardware on your phone for accurate predictions.

The Google Cloud Platform

Example: Automatic Video Stabilisation on the Pixel smartphone with Machine Learning.

Link on YouTube

Data sources as input for training the ML model:

- Video data (frames ordered by timestamps)
- Context data: series data on the camera's position and orientation from the on-board gyroscope and motion on the camera lens.

The Google Cloud Platform

For an eight-Megapixel camera:

- 8 million pixels per image $\times$ 3-channel layers $\simeq$ 23 million data points per image frame.
- 30 frames per second of video $\rightarrow$ quickly a short video becomes over a billion data points to feed into the model.

Conclusion:

We need a huge infrastructure to feed and train our ML models.

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Demo: Creating a VM on Computer Engine

- spin up a Compute Engine instance,
- run some software on it,
- copy files from that Compute Engine instance to cloud storage,
- publish those files to make them public.

Use case:

Earthquake activity from the U.S. Geological Survey website.

The Google Cloud Platform

Procedure:

1. Go to cloud.google.com
2. Create a new project in your billing account.
3. Select the *Compute Engine* section, go to *VM instances* and create a VM called "earthquake-vm". Enable first the API if asked. Pick a region and zone.
4. In the *Machine configuration* tab, choose an e2-micro instance and 10 GB of space.
5. In the *OS and storage* tab, change the default Debian image by an Ubuntu one.
6. In the *Networking* tab, allow http and https traffic.
7. In the *Security* tab, allow full access to all the cloud APIs, so that we can write to Cloud Storage from the VM.
8. Create the VM.
9. Access the VM console through SSH.
10. Install git.
11. Do `git clone https://www.github.com/GoogleCloudPlatform/training-data-analyst`.
12. Go into *courses/bdml_fundamentals/demos/earthquakevm*
13. Inspect the content of the `ingest.sh` and `transform.py` files.

The Google Cloud Platform

Procedure:

14. Run `install_missing.sh` to install the missing Python packages.
15. Run `ingest.sh` to download `earthquakes.csv`.
16. Inspect the beginning of `earthquakes.csv` with `head` and check the first few earthquake occurrences in the US.
17. Run `transform.py` to create an image (*earthquakes.png*) out of the csv file.
18. Create an *earthquake* bucket (`Storage/browser`) and verify with `gsutil ls gs://your-bucket-name` that it is empty.
19. Copy the produced data off the VM into cloud storage so that we can delete the VM.
20. Copy `earthquakes.*` to the bucket with `gsutil cp` and refresh the bucket in the GCP page.
21. Stop the VM (will be paused, I will keep paying for the storage but I will not pay for the computing) or delete it.
22. Make the data public so that we can access it from everywhere: go to *Permissions* and add members called *allUsers*, and give those users a *Storage Object Viewer* role.
23. Check the public links associated with our files and click on the htm's file link.

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Exercises:

Hands On!

1. Working with the Google Cloud Console and Cloud Shell
2. Working with Virtual Machines

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Kubernetes on the GCP

- We will learn how to set up and use Kubernetes and GCP's *Container Engine* to manage a set of machines and containers, and group them into services for hosting real world applications.

Clusters, Nodes and Pods

- Reminder:
  - A cluster is a set of computers...
  - managed by Kubernetes.
  - Kubernetes manages jobs.
  - Masters schedule jobs on nodes based on load or other criteria.
  - Jobs run containers on nodes.

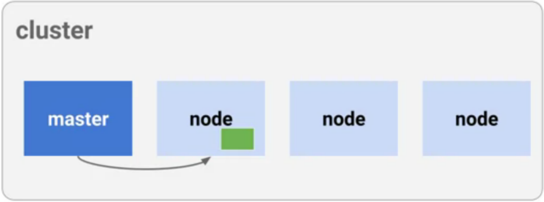

Image credits : Google

Clusters, Nodes and Pods

- A **Pod** is a type of job.
  - It can run one or more containers.
  - They share networking and storage separate from the node (they have their own IP).
  - Data is stored in volumes in memory or persistent disks
  - A **Pod** is defined with a YAML file.

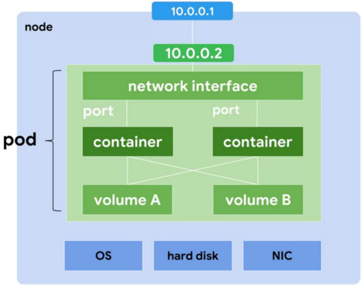

Image credits : Google

Clusters, Nodes and Pods

- A **Deployment** controller provides declarative updates for Pods and ReplicaSets.
- A desired state is described in a Deployment, and the Deployment controller changes the actual state to the desired state at a controlled rate.
- A deployment ensures that N pods are running in a cluster at a given time. If one of the pods goes down, Kubernetes has the ability to spin up another pod and replace it.
- A **Deployment** is defined with a YAML file.

Services

- A **Service** is an abstract way to expose an application running on a set of Pods as a network service.
- Kubernetes assigns a fixed IP and a single DNS name to the pod replicas and allows other pods or services to communicate with them.
- A **Service** is defined with a YAML file.
- There will be multiple services running with different configurations and features.
  - Usually, the frontend is a web server, application server, etc.
  - Backends are usually the nodes and maybe a database server.
  - They will usually apply a load balancer.

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Exercises:

Hands On!

1. Deploying Google Kubernetes Engine
2. Kubernetes Engine: Qwik Start
   - Enable the Kubernetes API in the Google Console
   - Check the Container registry for the Hello app: gcr.io/google-samples/hello-app
3. Running Dedicated Game Servers in Kubernetes Engine
4. Run & Build GameServer(Rust)- GKE+Agones Platform